

Instructions for Organizers

Introduction

Overview

The DataStorm problem statement is great for shorter events with participants that are new to MATLAB. It provides them with a weather dataset and challenges them with 10 data extraction and analysis problems. We suggest you familiarize yourself with the problems and potential solutions ahead of time, as participants may have questions.

Audience and Event Timing

We recommend the following time allotment for this event:

- 1-2 hours for an audience that may have experience with coding or MATLAB
- 2-3 hours for participants that are brand new to coding and MATLAB.

Pre-work

Pre-work is optional before this event, though you may choose to offer the following:

- [MATLAB Onramp](#)
- [Get familiar with MATLAB Documentation](#)

The Dataset

This problem statement uses a dataset consisting of climate data from 2000-2010. The dataset has 6 parameters:

1. AirTemperature – degree celcius
2. RainfallFlux - mm/day
3. SnowfallFlux - mm/day
4. SpecificHumidity - g/kg
5. SurfAirPressure - hPa
6. WindSpeed - m/s

Each parameter is stored as an 11x365 matrix. The rows represent years, the columns days, and the value at each (row,column) pair is the measured data at that day and year.

Name	Value	Size	Class
 AirTemperature	<i>11×365 double</i>	11×365	double
 RainfallFlux	<i>11×365 double</i>	11×365	double
 SnowfallFlux	<i>11×365 double</i>	11×365	double
 SpecificHumidity	<i>11×365 double</i>	11×365	double
 SurfAirPressure	<i>11×365 double</i>	11×365	double
 WindSpeed	<i>11×365 double</i>	11×365	double

Interested in using this problem statement?

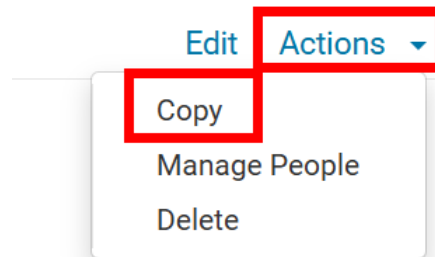
Reach out to hackathon@mathworks.com for access to the template materials in MATLAB Grader. You will need to meet the requirements for using MATLAB Grader:

- <https://www.mathworks.com/help/matlabgrader/ug/prerequisites-for-instructors.html>

Setup

Once you have access:

1. Go to <https://grader.mathworks.com/courses/167296>
2. Click on 'Actions' > 'Copy' on the right-hand corner

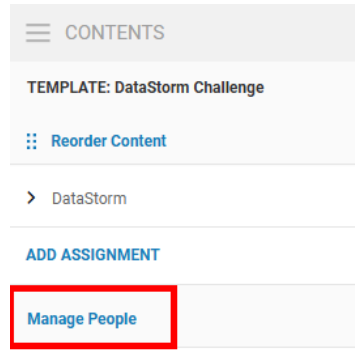


3. Click 'Copy' on the pop-up to create your own copy of this course, which should be called 'Copy of TEMPLATE: DataStorm Challenge' by default. This will open a new tab where you should do the following steps.
4. Name your event
5. Set the time zone
6. Set the start and end dates
7. Click 'Save'

Adding participants

Right before the event starts, you will need to add all participants to the MATLAB Grader course.

1. Go to grader.mathworks.com and click on the course you created earlier
2. Click on 'Manage People' on the left menu



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3. In the page that opens, select 'Learner' in the drop down to enroll learners, then copy and paste all participants' emails. You can do these all at once.

Manage People

A screenshot of the 'Manage People' dialog box. At the top right is an 'Enroll' button with a dropdown arrow. Below this, on the left, is a label 'Enroll' and a dropdown menu currently showing 'Learner'. This pair is highlighted with a red rectangle. To the right of the dropdown is a blue question mark icon. Below these is a large text input area containing the placeholder text 'participant1@email.com, participant2@email.com, participant3@email.com'. At the bottom right are 'Cancel' and 'Enroll' buttons.

If you are using Devpost, follow [these instructions](#) for getting a list of participant emails. During the welcome presentation, tell attendees to talk to you if they have not been added.

Evaluating

To export the results, click on the tab that says 'DataStorm':

A screenshot of the MATLAB Grader interface. The top header is 'MATLAB Grader'. Below it is a sidebar with 'CONTENTS' and a 'Close' button. The sidebar shows 'TEMPLATE: DataStorm Challenge', 'Reorder Content', and a collapsed 'DataStorm' item, which is highlighted with a red rectangle. Below the sidebar are 'ADD ASSIGNMENT' and 'Manage People' buttons. The main content area has tabs: 'Courses & Content', 'LMS Integration', and 'Documentation & Support'. The 'Courses & Content' tab is active, showing 'TEMPLATE: DataStorm Challenge'. It includes a duration '13 Jun 2025 - 30 Jun 2025', a list of products ('Optimization Toolbox, Statistics and Machine Learning Toolbox, Symbolic'), and a 'Course Description' section starting with 'Dive into the world of climate data analysis in MATLAB! Over the course of this'.

Then click on 'Actions' > 'Report' in the top-right corner

The screenshot shows the MATLAB Grader interface. On the left is a sidebar with a 'CONTENTS' menu and a list of challenges under 'Women in Tech Trek: DataStorm Challenge'. The main area displays the 'DataStorm' challenge details, including its due date (29 Nov 2025 1:30 PM EST) and an 'Assignment Description'. In the top right corner, an 'Actions' dropdown menu is open, with the 'Report' option highlighted. Other options in the menu include 'Edit', 'Copy', and 'Delete'.

On the popup that opens, choose to export the Best Solutions as of the due date as a CSV, then click 'Report'. This should download a .CSV file called 'DataStorm'.

Create an Assignment Report ✕

Report includes, for each learner and assessment item:

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Best solution as of the due date (29 Nov 2025 1:30 PM EST)

☐

Best solution as of today☐

Output Format:

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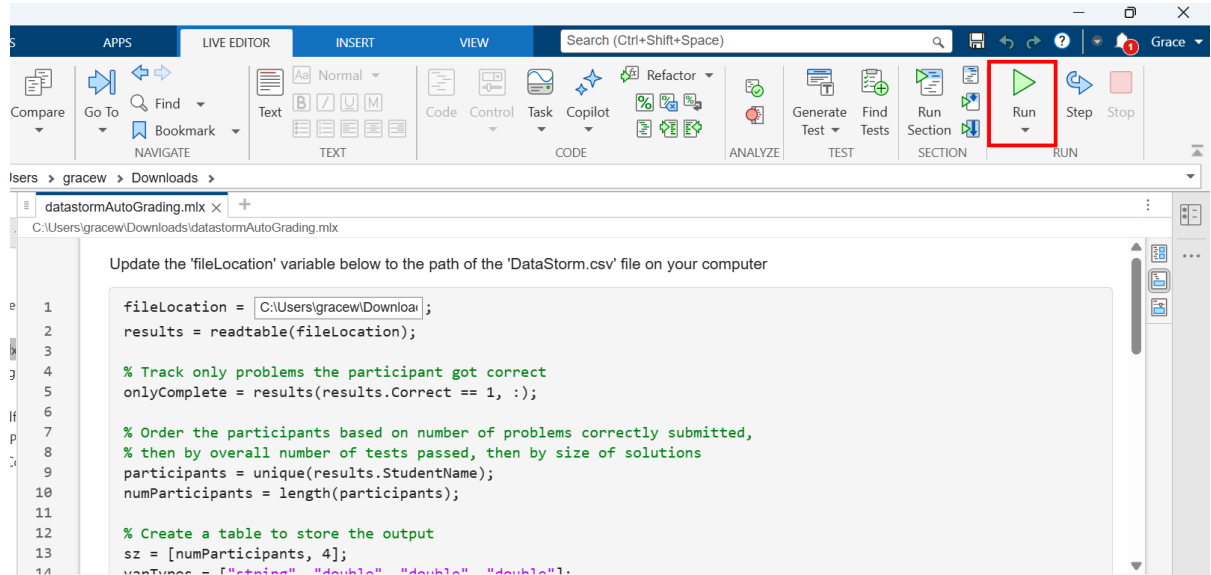
CSV☐☐

Cancel

Report

Use this file with the MATLAB Live Script 'datastormAutoGrading.mlx' to determine the winners:

1. Open MATLAB
2. Open 'datastormAutoGrading.mlx'
3. Update the fileLocation to be the file path of the results (ex:
C:\Users\gracew\Downloads\DataStorm.csv)
4. Run the live script by clicking on the 'Run' Button



5. Scroll to the bottom and use the output to determine the winning participants/teams. This should be the first, second, and third rows of the table.